

ECD 415 PROJECT PHASE 1

PRESENTED BY: GROUP 01

AMAYA PRAMOD	KNR23EC013
HARIKESH OP	KNR23EC044
NANDANA JAYACHANDRAN	KNR23EC064
POOJA K	LKNR23EC111

GUIDED BY:
DR. SAJESH KUMAR U

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
GOVERNMENT COLLEGE OF ENGINEERING KANNUR

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OUTLINE

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Topic 1: Edge AI Smart Battery Management System with Intelligent UPS

INTRODUCTION

Lithium-ion battery packs are widely used in EVs, renewable energy storage, robotics, and industrial backup systems.

Conventional BMS provides basic protection (over/under-voltage and over-current) with limited battery health insight.

The proposed Intelligent BMS integrates Edge AI, predictive analytics, and UPS capability on an ESP32-S3 platform.

Enables real-time battery intelligence through SOC, SOH, RUL estimation, and on-device fault detection.

PROBLEM STATEMENT

Conventional BMS acts mainly as a protection device.

It cannot accurately estimate battery health, predict degradation, or Remaining Useful Life (RUL).

This may lead to premature or delayed battery replacement, increasing cost and reducing reliability.

Need: An Edge AI-based Intelligent BMS for predictive analytics, real-time monitoring, and intelligent UPS management.

Conventional Battery Management Systems lack the predictive intelligence required for reliable, efficient, and proactive battery management.

PROPOSED METHODOLOGY

Acquire battery data – Monitor cell voltage, current, and temperature using embedded sensors.

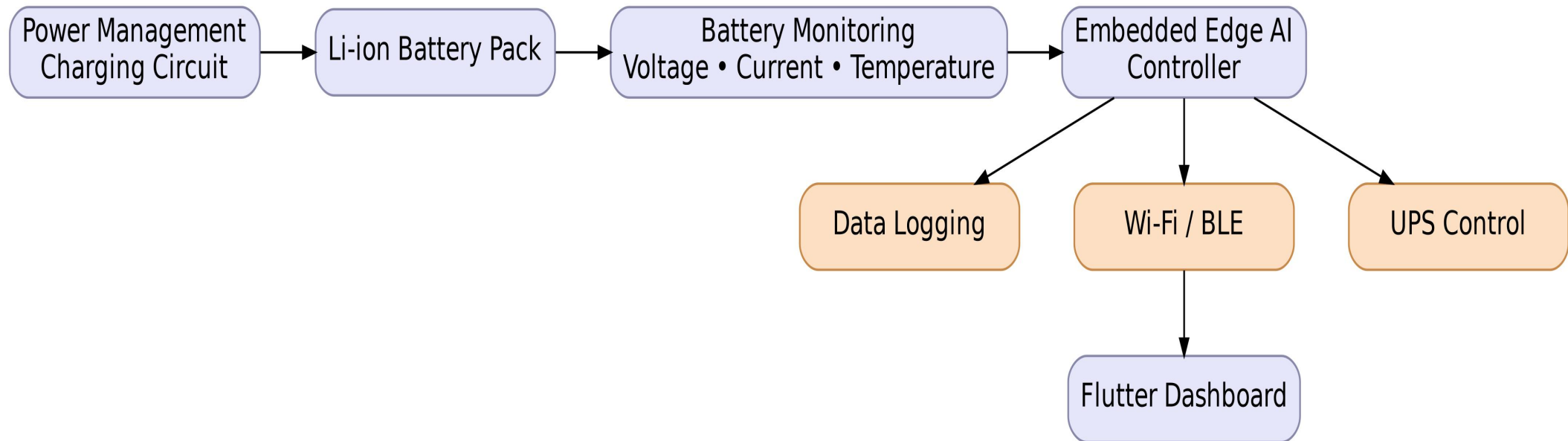
Battery monitoring – Estimate battery parameters and execute protection algorithms.

Edge AI model – Train a lightweight ML model using historical battery data to predict degradation, SOH, RUL, and faults.

On-device inference – Deploy the trained model on ESP32-S3 for real-time prediction without cloud dependency.

Smart battery management – Intelligent UPS switching, wireless battery monitoring, and Flutter dashboard visualization.

BLOCK DIAGRAM



EXPECTED OUTCOMES

- Functional Edge AI Smart Battery Management System.
- Real-time monitoring of battery parameters.
- Accurate SOC and SOH estimation.
- Remaining Useful Life prediction.
- Intelligent battery fault detection.
- Automatic UPS switching.
- Flutter-based monitoring dashboard.
- Embedded AI inference without cloud dependency.

APPLICATION

- Electric Vehicles.
- Renewable Energy Storage Systems.
- Smart UPS Systems.
- Robotics.
- Industrial Battery Monitoring.
- Telecom Backup Systems.
- Medical Backup Power.
- Predictive Maintenance Platforms.

Topic 2: Embedded Electronics Diagnostic Instrument

INTRODUCTION

This project proposes a portable diagnostic instrument that combines multiple electronic testing functions on a single embedded platform.

It integrates oscilloscope functionality, logic analysis, and protocol decoding for UART, SPI, I2C, and CAN.

The instrument also supports waveform visualization, signal measurement, and data logging, with wireless connectivity to a Flutter dashboard.

PROBLEM STATEMENT

Existing electronic diagnostic equipment is expensive, bulky, and often limited to laboratory environments.

Students, researchers, and field engineers require an affordable portable diagnostic instrument capable of analyzing multiple electronic interfaces using a single embedded platform.

The lack of affordable and portable diagnostic solutions limits efficient electronic system analysis beyond laboratory environments.

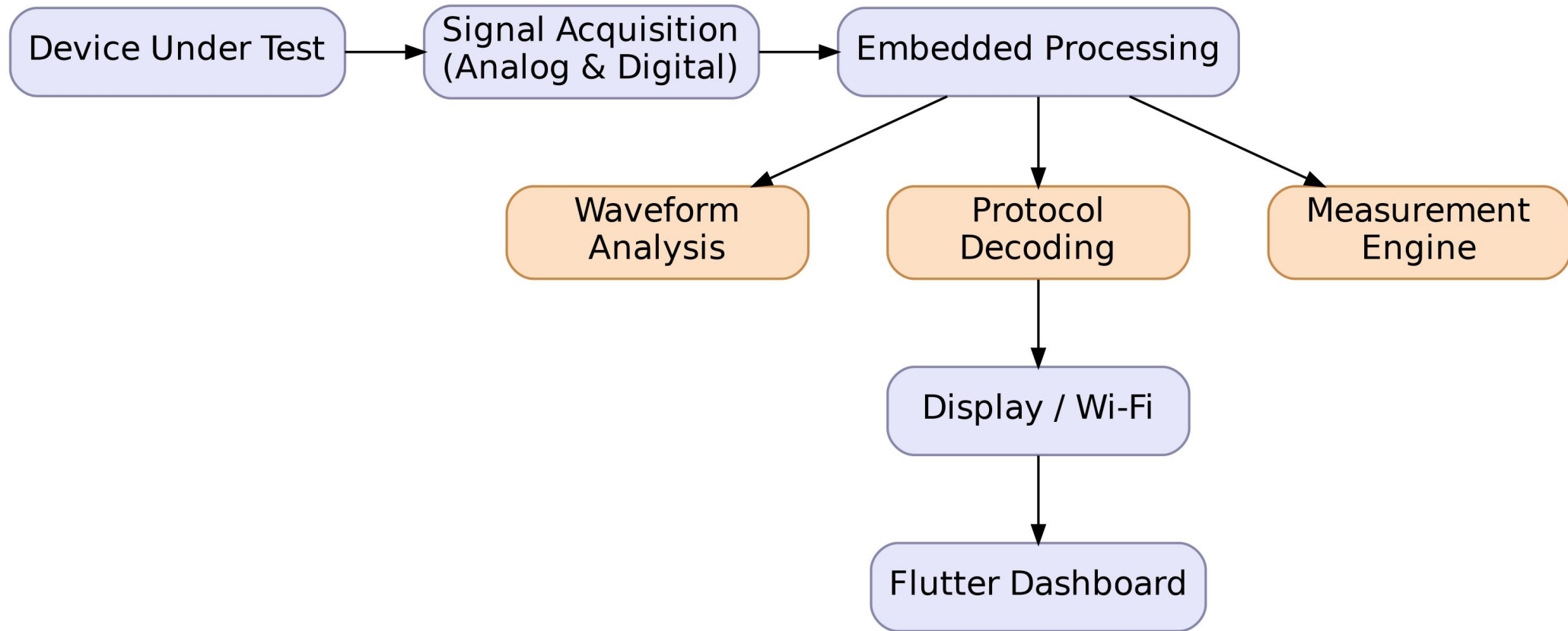
PROPOSED METHODOLOGY

Signal acquisition circuits capture analog and digital signals from the Device Under Test.

The embedded processor performs waveform sampling, protocol decoding, and signal analysis.

Captured information is displayed locally and transmitted wirelessly to a Flutter dashboard for visualization and storage.

BLOCK DIAGRAM



EXPECTED OUTCOMES

- Portable diagnostic instrument.
- Oscilloscope functionality.
- Logic analyzer.
- Protocol decoder.
- Wireless monitoring.
- Data logging.

APPLICATION

- Embedded System Development.
- PCB Debugging.
- Industrial Maintenance.
- Laboratory Education.
- Robotics.
- Automotive Electronics.

Topic 3:

Adaptive Lora Mesh Emergency Communication Network

INTRODUCTION

This project proposes an Adaptive LoRa Mesh Emergency Communication Network designed to provide reliable long-range communication when conventional communication infrastructure becomes unavailable.

The system consists of intelligent LoRa nodes capable of self-organizing into a mesh network, dynamically routing messages, and maintaining communication during disasters.

Wireless connectivity, GPS integration, mobile application support, and intelligent routing algorithms enable reliable communication in emergency situations.

PROBLEM STATEMENT

Existing emergency communication systems depend heavily on cellular networks and internet connectivity.

During disasters, these infrastructures may become unavailable, preventing effective coordination among rescue teams.

There is a need for a portable, self-organizing, and energy-efficient emergency communication system.

Ensuring reliable communication in the absence of conventional network infrastructure remains a critical challenge.

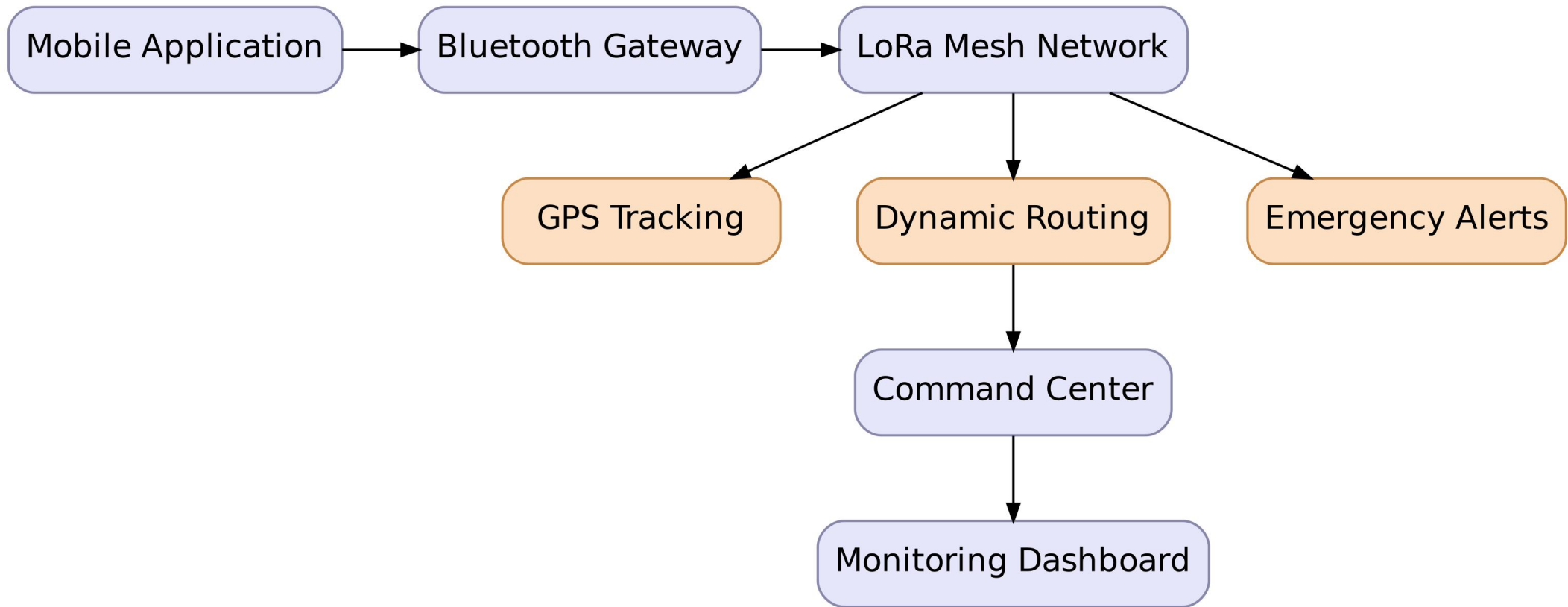
PROPOSED METHODOLOGY

Each communication node continuously exchanges routing information with neighboring nodes to establish a self-healing mesh network.

Messages are dynamically routed through the optimal communication path.

GPS coordinates, emergency alerts, and sensor information are transmitted between nodes and displayed on a Flutter application and monitoring dashboard.

BLOCK DIAGRAM



EXPECTED OUTCOMES

- Functional LoRa Mesh Network.
- Self-healing routing.
- Long-range communication.
- GPS location tracking.
- Flutter dashboard.
- Emergency alert system.

APPLICATION

- Disaster Management.
- Search and Rescue.
- Forest Monitoring.
- Military Communication.
- Smart Agriculture.
- Remote Area Communication.